CYBR-2200 Network Analysis

WEEK 2 — Identifying Hosts, Devices, and Users

📊 Gradebook Configuration

- **Assignment Name:** Week 2 Lab: Identifying Hosts and Users
- **Category:** Weekly Labs
- **Point Value: 90 points**
- **Counts Toward:** Weekly attendance & engagement
- **Submission Type:** File upload (DOCX or PDF)
- **GitHub:** ❌ Not used this week

🎯 Assignment Purpose

This lab teaches you how security analysts identify:

- hosts
- devices
- operating systems
- users (when visible)

using **network traffic alone**.
You are not expected to analyze malware.

📝 What Students Must Do (Explicit)

Students must analyze **all PCAP files provided for Week 2** and complete **every section** of the worksheet:

Part A — Open PCAP Files

- Open each PCAP successfully in Wireshark

Part B — Host Identification

- Identify IP addresses
- Locate hostnames (when present)
- Identify MAC addresses or vendors
- Note which protocol revealed the information

2022-MTA-workshop-Win11-host.pcap

IPv4/IPv6	Hostname	MAC Address	Protocol
192.168.42.170/ fe80::f5d4:82f0:5808:5db2	DESKTOP-WIN11PCQ	00:16:eb:f8:1c:c8 (Intel_f8:1c:c8)	DHCP

OS/Device	Protocol	User Account	Reason
Windows 11	DHCP	None Found	None needed

Windows 11 PCAP: Source IP of packet 8 is 192.168.42.170. The IPv6 of the source is fe80::f5d4:82f0:5808:5db2 from packet 12. The hostname is DESKTOP-WIN11PCQ. The MAC address is 00:16:eb:f8:1c:c8 (Intel_f8:1c:c8). DHCP revealed the hostname, IP address, and MAC address. The DHCP parameter request list and vendor identifiers indicated a Windows operating system.

2022-MTA-workshop-IPhone.pcap

IPv4/IPv6	Hostname	MAC Address	Protocol
10.8.21.213	Brad's iPhone	b0:ca:68:d9:19:82	DHCP

OS/Device	Protocol	User Account	Reason
iPhone OS 15.6.1	HTTP	None Found	Not needed

iPhone PCAP: Using the HTTP and DHCP packets, we determined this was Brad's iPhone running OS 15.6.1. The phone's MAC address is b0:ca:68:d9:19:82. No user accounts were found to be extracted either due to encryption or not needed.

2022-MTA-workshop-Fedora-Linunx.pcap

IPv4/IPv6	Hostname	MAC Address	Protocol
10.8.30.128	fedoraproject.org	00:0c:29:53:51:c0	HTTP
OS/Device	**Protocol**	**User Account**	**Reason**
Apache Server using Fedora as a proxy server	HTTP stream	X-Fedora-ProxyServer proxy03.fedoraproject.org	**By using a proxy server, another layer of security helps the Fedora user secure their information.**

Fedora PCAP: Using the HTTP and TCP packets, we determined this was a Fedora proxy server for an Apache server. The Linux user's MAC address is 00:0c:29:53:51:c0**. The user account we were able to extract via TCP and HTTP stream following is the** X-Fedora-ProxyServer proxy03.fedoraproject.org**.**

2022-MTA-workshop-Android-phone.pcap

IPv4/IPv6	Hostname	MAC Address	Protocol
172.16.1.121/fe80::f09a:83ff:fe00:897d	**Galaxy-S20-FE-5G**	**f2:9a:83:00:89:7d**	**DHCP**
OS/Device	**Protocol**	**User Account**	**Reason**
Samsung Galaxy S20	**DHCP**	**None found or encrypted**	**Encrypted**

2022-MTA-workshop-macbook.pcap

IPv4/IPv6	Hostname	MAC Address	Protocol
10.8.22.101	**Brad-MBP.local**	**a4:83:e7:2a:45:d7**	**DHCP**
OS/Device	**Protocol**	**User Account**	**Reason**
macOS	**MDNS**	**None found**	**This was shown in the MDNS packets when followed.**

Part C — OS & Device Indicator

- Identify operating system clues
- Explain which protocols revealed them

Windows 11 PCAP: The Windows 11 host was identified through DHCP traffic. The IPv4 address 192.168.42.170 and IPv6 address fe80::f5d4:82f0:5808:5db2 were observed in DHCP exchanges. The hostname DESKTOP-WIN11PCQ was disclosed in the DHCP request, along with MAC address 00:16:eb:f8:1c:c8 (Intel vendor). DHCP option fields and vendor identifiers were consistent with a Windows 11 operating system.

iPhone PCAP: The device in this packet capture was identified as an Apple iPhone "Brad's iPhone" through DHCP and HTTP traffic analysis. The DHCP Request packet revealed the MAC address b0:ca:68:d9:19:82, which determines Apple product based on the Organizationally Unique Identifier (OUI). The DHCP exchange also disclosed the hostname "Brad's iPhone" under Option 12 (Host Name). The device was assigned the IPv4 address 10.8.21.213 by the DHCP server at 10.8.21.1, which appears to be the local gateway. Extra evidence produced confirmation of the operating system was obtained through HTTP traffic analysis. The HTTP User-Agent string contained the identifier:
Mozilla/5.0 (iPhone; CPU iPhone OS 15_6_1 like Mac OS X)
This confirms the device was running iOS version 15.6.1. No user account credentials or authentication exchanges were detectable in the capture. The traffic mostly consisted of DHCP communication and encrypted HTTP activity. No usernames or passwords were observed due to the traffic being encrypted in this PCAP.

Android: The device in this capture was identified as a Samsung Galaxy S20 FE 5G running Android 12 through DHCP analysis. The DHCP Discover and Request packets revealed the vendor identifier string "android-dhcp-12," indicating an Android 12 operating system. The hostname visible within the DHCP payload included "Galaxy-S20-FE-5G7," identifying the device model.

The device was assigned the IPv4 address 172.16.1.121 by the DHCP server at 172.16.1.1. The MAC address observed was f2:9a:83:00:89:7d, which appears as a locally administered (randomized) address, a common privacy feature in modern Android devices.
No user authentication exchanges or credential information were visible in the capture. The traffic consisted primarily of DHCP communication, and no cleartext user account information was exposed.

Macbook Pcap: The device in this capture was identified as a MacBook running macOS through DHCP and mDNS analysis. The DHCP Request packet revealed the MAC address a4:83:e7:2a:45:d7, which resolves to Apple, Inc. The device was assigned the IPv4 address 10.8.22.101 by the gateway 10.8.22.1.
Multicast DNS (mDNS) traffic advertised the hostname Brad-MBP.local, a naming convention typical of macOS systems. Service advertisements such as _ssh._tcp.local further support macOS identification.
HTTP traffic included a request for /hotspot-detect.html, a standard macOS internet connectivity check. Although the exact macOS version was not visible, the Apple OUI, hostname structure, and Bonjour service behavior strongly indicate a MacBook system.
No user credentials or authentication exchanges were observed. The traffic consisted primarily of DHCP, ARP, mDNS, and standard HTTP activity.

Part D — User Identification

- Identify user account info (if present)
- Explain why user info may not be visible

In the reviewed packet captures, no user account credentials were visible within any of the PCAPS. The traffic primarily consisted of DHCP, ARP, mDNS, and encrypted HTTP communications. Usernames and passwords are typically visible only during authentication exchanges, such as unencrypted login submissions or cleartext application-layer protocols. Because these captures did not include those protocols, no user account information was able to be recovered.
This highlights the importance of understanding both what is visible in network traffic and what is protected by encryption. Personal identifiable information is typically encrypted. While host identification is typically possible through DHCP and mDNS protocols, user-level designation more direct observation such as social engineering or correlating logs from other sources.

Part E — Analyst Reflection

- Explain why host and user identification matters

Host and user identification are the most basic elements of network analysis and incident response. Identifying a device allows analysts to assign network activity to a specific endpoint, detect unauthorized systems, and establish baselines for normal behavior. Without accurate host identification, it is difficult to determine whether activity is legitimate or suspicious.

During investigations involving policy violations, account compromise, or lateral movement, user identification becomes significant. While encryption protects credentials and PII, analysts must rely on behavioral patterns and contextual evidence to connect activity with specific users. The ability to identify hosts on a network and determine whether user data is available are necessary skills needed for professional packet analysis and strengthens response time effectiveness and incident response effectiveness.

📤 Submission Requirements

Students submit:

- **One completed worksheet**
- DOCX or PDF format
- Uploaded to Blackboard

🚫 No PCAP uploads
🚫 Screenshots optional only